

Section 10.5

1. Find the derivative $g'(t)$ of the given function $g(t)$.

(a) 10 points $g(t) = \frac{4}{\sqrt{t}}$

First, we re-write $g(t)$ as

$$g(t) = 4t^{-1/2}.$$

Then, we apply the power rule to yield

$$g'(t) = (4) \left(-\frac{1}{2} t^{-(1/2)-1} \right) = -2t^{-3/2}.$$

(b) 10 points $g(t) = 4 + 4t + 4t^2 + 4t + 4$

We apply the power rule to yield

$$g'(t) = 4 + (4)2t + 4 = 8 + 8t.$$

(c) 10 points $g(t) = (t + 1)^3 + 9$

First, we expand $g(t)$ to yield

$$g(t) = t^3 + 3t^2 + 3t + 10.$$

We then apply the power rule to yield

$$g'(t) = 3t^2 + (3)2t + 3 = 3t^2 + 6t + 3.$$

2. 10 points Let f and g be functions satisfying $f'(3) = 7$ and $g'(3) = 1$. If $h(x) = f(x) - 2g(x) + 2$, find $h'(3)$.

We use the linearity of the derivative to yield

$$h'(x) = f'(x) - 2g'(x).$$

Then, evaluating at $x = 3$, we use the given information that $f'(3) = 7$ and $g'(3) = 1$ to obtain

$$h'(3) = f'(3) - 2g'(3) = 7 - 2(1) = 5.$$

3. 10 points A company's cost (in dollars) for producing x desks is given by

$$C(x) = 100x - 10x^2 + x^3.$$

Find $C(5)$ and $C'(5)$. Show your work. Write a brief, informal interpretation of your results.

First, we calculate $C(5)$ and yield

$$C(5) = 100(5) - 10(5)^2 + (5)^3 = 500 - 250 + 125 = 375.$$

This tells us that it costs \$375 for the company to produce 5 desks. Then, we calculate $C'(x)$ by using the chain rule to obtain

$$C'(x) = 100 - (10)2x + 3x^2 = 100 - 20x + 3x^2.$$

Following this, we calculate $C'(5)$ and yield

$$C'(5) = 100 - 20(5) + 3(5)^2 = 100 - 100 + 3(25) = 75.$$

This tells us that the instantaneous rate of change in the cost of production when producing 5 desks is \$75. That is, the total cost will increase by approximately \$75 to produce another desk (that is, to produce 6 desks).

4. 10 points (bonus) Let $f(x) = x^4$ and let $g(x) = x^2 + 2$. Using the fact that the correct derivative of $f(g(x))$ is given by

$$(f(g(x)))' = 8x(x^2 + 2)^3,$$

which one of the following statements is true?

- (a) $(f(g(x)))' = f(g'(x))f'(x)$
- (b) $(f(g(x)))' = f'(g(x))g'(x)$
- (c) $(f(g(x)))' = f'(x)g'(x)$

Hint: You don't need to expand $8x(x^2 + 2)^3$ to get the correct answer.

As we know the correct derivative, we simply compute each option and check to see if it is correct. Deriving f and g , we get

$$f'(x) = 4x^3 \quad \text{and} \quad g'(x) = 2x.$$

Option (a) yields

$$f(g'(x))f'(x) = (2x)^4(4x^3) = 64x^7 \neq 8x(x^2 + 2)^3.$$

So, option (a) is not correct. Option (b) yields

$$f'(g(x))g'(x) = 4(x^2 + 2)^3(2x) = 8x(x^2 + 2)^3.$$

So, option (b) is correct. If we were to calculate option (c), we would yield

$$f'(x)g'(x) = (4x^3)(2x) = 8x^4 \neq 8x(x^2 + 2)^3.$$

So, option (c) is not correct either.

Section 10.7

1. The price-demand equation for the production of bicycles is given by the following equation, where x is the quantity of bicycles demanded and p is price (in dollars)

$$x = 5000 - 25p.$$

- (a) 15 points Express the revenue $R(x)$ as a function of the demand x .

Hint: revenue equals price times demand.

First, we manipulate the price-demand equation so that p is expressed as a function of x , yielding

$$p = -\frac{1}{25}x + \frac{5000}{25} = -\frac{1}{25}x + 200.$$

Then, we note that revenue is equal to price times demand, so we need only multiply our expression for p (in terms of x) by x to obtain the revenue function

$$R(x) = x \left(-\frac{1}{25}x + 200 \right) = -\frac{1}{25}x^2 + 200x.$$

- (b) 10 points Find the marginal revenue $R'(x)$.

Using the revenue function

$$R(x) = -\frac{1}{25}x^2 + 200x$$

we calculate $R'(x)$ by the power rule to yield

$$R'(x) = -\frac{2}{25}x + 200.$$

- (c) 10 points Find $R(2500)$ and $R'(2500)$, and give a brief, informal interpretation of your results.

Using the revenue function

$$R(x) = -\frac{1}{25}x^2 + 200x$$

and the marginal revenue function

$$R'(x) = -\frac{2}{25}x + 200,$$

we calculate $R(2500)$ and $R'(2500)$

$$R(2500) = -\frac{1}{25}(2500)^2 + 200(2500) = 250,000$$

$$R'(2500) = -\frac{2}{25}(2500) + 200 = 0.$$

This tells us that when the quantity of bicycles demanded is 2500, we yield a revenue of \$250,000, and the marginal revenue at this level is 0.

As the revenue function is a parabola opening downward, the only tangent line with zero slope occurs at the vertex – which achieves the largest y value for the function (that is, the largest revenue). Thus, this means that \$250,000 is the maximum revenue the firm can yield. This does not necessarily mean that producing 2500 bicycles is profit-maximizing level of production, however. To know that, we would need to know the marginal cost function. Then, if we did know the marginal cost function, and if marginal cost equals marginal revenue at a production level of 2500 bicycles, it would then be the profit-maximizing level of production.

2. 15 points If the cost of producing x cell phones is given by $C(x) = 300x + 100$, find the marginal average cost of producing x phones.

Using the cost function, we compute the average cost function

$$\bar{C}(x) = \frac{C(x)}{x} = \frac{300x + 100}{x} = 300 + 100x^{-1}.$$

Using the power rule, we differentiate $\bar{C}(x)$ to yield

$$\bar{C}'(x) = -100x^{-2}.$$